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Review of short-text classification

Short-text
classification

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Abstract

Purpose – Rapid developments in social networks and their usage in everyday life have caused an explosion in the amount of short electronic documents. Thus, the need to classify this type of document based on their content has a significant implication in many applications. The need to classify these documents in relevant classes according to their text contents should be interested in many practical reasons. Short-text classification is an essential step in many applications, such as spam filtering, sentiment analysis, Twitter personalization, customer review and many other applications related to social networks. Reviews on short text and its application are limited. Thus, this paper aims to discuss the characteristics of short text, its challenges and difficulties in classification. The paper attempt to introduce all stages in principle classification, the technique used in each stage and the possible development trend in each stage.

Design/methodology/approach – The paper as a review of the main aspect of short-text classification. The paper is structured based on the classification task stage.

Findings – This paper discusses related issues and approaches to these problems. Further research could be conducted to address the challenges in short texts and avoid poor accuracy in classification. Problems in low performance can be solved by using optimized solutions, such as genetic algorithms that are powerful in enhancing the quality of selected features. Soft computing solution has a fuzzy logic that makes short-text problems a promising area of research.

Originality/value – Using a powerful short-text classification method significantly affects many applications in terms of efficiency enhancement. Current solutions still have low performance, implying the need for improvement. This paper discusses related issues and approaches to these problems.

Keywords Social networks, Classification, Sentiment analysis, Feature selection, Short text

Paper type Literature review

1. Introduction

The growth of social networks has provided people new ways to access information through the internet. As an alternative to traditional online sources, such as (websites) that have previously represented the sources of information, microblogs nowadays provide a valuable information portal on different topics ranging from politics to entertainment. Considering that the user can select the type of information that they are interested in and given its speed, efficiency and comprehensive characteristic, people increasingly tend to use social networks, such as Twitter, Google+ and Facebook, in their life. Millions of short text are produced daily in the form of posts or comments. This type of document tends to have a length of no more than 200 characters; for example, Twitter posts consists of up to 140 characters.

However, users of social networks have different interests and preferences. Many people face a problem in finding appropriate data at a reasonable time. Thus, they need a way to



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identify the most relevant data and to classify them based on the topics covered or a set of other feature, such as the follower user. As a result, text classification is becoming important in many problems, such as in sentiment analysis, tweet personalization and spam filtering. The classification of short texts uses the same series of steps as when classifying long texts. This process involves a feature extraction step and a classification step, which classifies text by using the information of labeled data and features from training data. Compared with the classification of the long texts, the main difficulty in short-text classification is the brevity of the document and sparsity of the feature space. The general framework of this classification process is summarized in [Figure 1](#).

Researchers have recently made various attempts to cope with the problem of the classification of short texts in microblogs. [Lee et al. \(2011\)](#) proposed two approaches for classifying trending topics in Twitter in a set of predefined classes. The first approach depends on the bag-of-words approach and uses a multinomial naïve Bayes algorithm to classify tweets based on their text content. The second approach is network based and uses the C5.0 decision-tree (DT) classifier. This approach takes advantage of the social network structure of Twitter, such as using the number of active mutual users engaged in the provided topic. The experimental results revealed that the second approach outperformed the first one with F1 measures of 77 and 70 per cent, respectively.

[Irani et al. \(2010\)](#) proposed a method for classifying tweets to the corresponding trends, improving the categorization performance by combining the tweet with its associated web page and using the information obtained for the reduction of the feature space. [Fiaidhi et al. \(2013\)](#) proposed a hierarchical ensemble-classification approach for trend topics. KNN, TF_IDF, NB and LANGUAGE MODEL were combined to classify each tweet to its relevant class. The data set was built using the T3C to collect trending topics, which were then labeled. The experiment shows that ensemble classification achieved about 75 per cent accuracy and outperformed the other classifiers when it is used alone. Moreover, the accuracy of the classifier was increased by using the language model with N-gram. [Weissbock et al. \(2013\)](#) attempted to solve the problem of shortness of the texts by expanding the tweets that contain links to access a webpage. To increase the number of words, a tweet was extended by appending the title of the web page, as well as the top 10 frequently used terms in the web page. The evaluation of an experiment using three familiar classifiers, namely, the support vector machines (SVM), naïve Bayes (NB) and DT, demonstrated that expanding the tweet with external data enhances the classification accuracy. [Wang et al. \(2013\)](#) proposed a feature selection approach that exploits the data content of the short text itself, regardless of any external data. The feature to be selected must appear in a sufficient number of documents, and it should be representative of the topic. By separating the feature selection from the feature vector construction, the proposed method achieved positive results with a web snippet data set using the NB classifier. However, this approach took a long time to accomplish the classification. [Cotelo et al. \(2016\)](#)

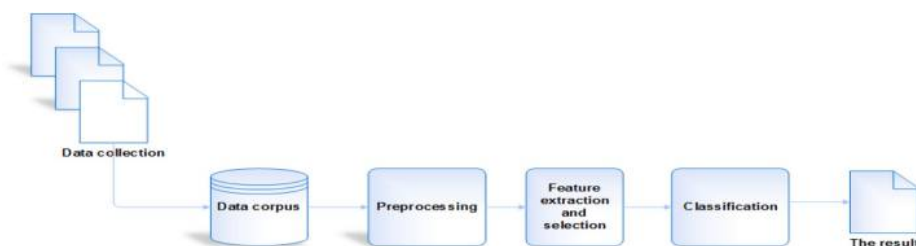


Figure 1.
The classification process

propose a tweet classification approach that combines both structural information and textual contents. The experiments show that the mixed approach surpasses the state-of-art methods. [Demirsoz and Ozcan \(2016\)](#) used a binary classification to detect tweets that related to a news article. The tweets were classified using a textual similarity-based method. The similarity measure computes the probability of a tweet to belong to the relevant set of tweets that related to the certain news article.

This paper is structured as follows. Section 2 presents the challenges to short texts. Section 3 presents an overview of short-text classification. Section 4 presents the preliminary process of short-text classification. Section 5 describes the feature selection approaches, whereas Section 6 illustrates the feature extraction approach. Section 7 discusses machine-learning classifiers. Section 8 presents the classification performance measures. Section 9 concludes the paper.

2. Challenges to short texts

Short texts involve the type of problem that deals with documents that are relatively short in length. A wide range of applications utilize this kind of text, such as microblogs, Twitter, mobile messages and news comments. The length of these texts is extremely short and usually has a maximum of 200 characters. For instance, tweets are no more than 140 characters, while a short mobile message is less than 70 characters. According to [Song et al. \(2014\)](#), all short texts have the following attributes:

- *Shortness*: A short text contains a few words, which may result in an inadequate representation of the document.
- *Sparsity*: A short text has a limited length. This limited capacity should be used to express many different topics with users each using his/her own words and writing style. Therefore, a certain topic will have diverse content; it is challenging to obtain its precise features.
- *Instant*: Considering its application, a short text is processed and transmitted in real time.
- *Misspelling and informal writing*: In most cases, especially in comments in microblogs, a short text is presented briefly, including many misspellings and noise and the existence of a particular language.

According to [Faguo et al. \(2010\)](#), sparsity and shortness significantly affect the performance of machine-learning classifiers. Given that traditional machine-learning methods cannot be used in short texts, the word content of a short text may be highly diverse while also having limited number of words in each post. This condition complicates the feature space construction for text categorization. Feature space augmentation, specifically, feature enrichment, is one of the most efficient solutions for coping with the problem of shortness. According to [Kamath and Caverlee \(2011\)](#), three approaches for feature enrichment are available:

- (1) *External-based enrichment*: This technique adds features obtained from an external source to increase the number of words in a feature set, such as by using the link inside a post in a microblog or through collecting the features from the web page associated with these links. We can also enrich a short text depending on the context of the short-text corpus, semantic similarity or through the search engine to find web pages from trusted sites. After web pages are collected, the terms found can be directly added to the short-text corpus. Others can depend on topic taxonomy, for example, by using a Wikipedia category to enrich the short text.

Another method is by using the parts of speech (POS), for instance, recognizing nouns in a short text usually provide a reliable understanding of the whole message.

- (2) *Lexicon-based enrichment*: To solve the sparsity problem of terms, we can use a lexical approach. In character-based n-grams, the feature set is formed by taking a string of m adjacent characters in the document. On the other hand, in word-based n-grams, the feature set is built from the words in the document.
- (3) *Collocation-based enrichment*: [Manning \(1999\)](#) stated that collocation implies that “two or more words together form an expression that matches the way saying things”. Collocation can add additional features to the feature space. The most important factor in collocation is the association measure, which is a mathematical method that is used to measure how much the words in a phrase are related to each other. Essentially, this method estimates the co-occurrences between the words in the phrase. Examples of this measure are mutual information, log-likelihood ratio and chi-squared methods.

3. Preprocessing and their effect on classification

As previously, text classification is usually done through sequential steps starting by applying several purification operations on real data to remove all impurities and represent all of the data in standard form to ensure that they are ready to be used in different machine-learning algorithms. The process of cleansing the text and removing its impurities is referred to as the preprocessing step that can alter the performance of the categorization task ([Haddi et al., 2013](#)). This step is followed by choosing the most discriminative term in the text. Tokenization, stemming, stop-word removal and lowercasing are considered standard preprocessing operations shared by most text-classification algorithms.

Tokenization involves splitting a text into a meaningful piece. Common words that do not benefit the categorization process, which are referred to as stop words, are then removed. Therefore, the space size is increased, which delays the whole system. Stemming is the process of identifying the root of words. Stemming is intended to find the stem of a group of words that are derived from this stem. The last operation is to convert the letters of all words to lowercase letters. The classification of data collected from microblogs requires an additional process; a specific method must deal with the informality and spelling mistakes that frequently appear in this type of documents. Various preprocessing operations depend on the nature and application of the collected data. However, all operations were used to ensure that the data do not increase the dimension of the feature space and reduce the performance and complexity of the classification task.

4. Weighting schemes in short-text classification

To classify a text or obtain information from it, determining which words are significant within the text is needed. The most popular way to accomplish this task is by assigning a numeric value to each word to reflect its contribution to document classification. This value is referred to as the weight of the term. The TF-IDF scheme is a weighting scheme that is extensively used today because of its simplicity and efficiency in classification.

To classify a text, a document should be represented as a set of terms; the term itself can be single or have multiple words. Single-word terms are generally used in text representation. Multi-word terms can be one of the following: syntactic phrases, statistical phrases and term sets. Syntactic phrases ([Scott and Matwin, 1999](#)) are a concatenation of

words arranged by syntactic relations. Familiar phrases are usually verb, noun and adjective phrases. Statistical phrases (as n-grams) (Bekkerman and Allan, 2003) are a series of n contiguous words that are used to define co-occurrence-based features. A term set (Badawi and Altınçay, 2014) is a sequence of words in which the co-occurrences of the terms are not necessarily adjacent.

The term weights are calculated based on the statistical information of words within the document, or semantic weights. The semantic weights scheme utilizes the semantics of classes for indexing. Luo *et al.* (2011) stated that the semantics of the class are represented by the senses of terms occurring in the class labels, as well as the analysis of the terms by WordNet.

Weighting schemes are divided into two categories based on whether or not the scheme requires the class information of training documents in the classification process. Unsupervised weighting schemes do not use class information to discriminate the term, such as TF, TF-IDF, and its variants. On the other hand, supervised weighting schemes use class information, such as TF*IG and TF*CHI2. Many researchers (Martineau *et al.*, 2008; Ren and Sohrab, 2013) proposed new term-weighting schemes for text classification, and each of the researchers considered a different measure to express the importance of the term within a document.

4.1 Unsupervised term-weighting method (traditional)

Traditionally, text classification uses binary weight that represents the simplest method, or the TF and the term-weighting schemes of its variants. The TF*IDF is considered the most common scheme for weighting because of its simplicity and efficiency. Table I summarizes the most common and traditional term frequency schemes.

4.2 Supervised term-weighting method

Supervised term weighting was derived from supervised learning that traditionally affects classification and term selection. Thus, the weight reveals the importance of a particular term, which can determine whether a document belongs to a particular category or not, by using information on the membership in a training corpus. To illustrate supervised term weighting, Erenel *et al.* (2011) integrated the TF-IDF and the set of selection function by replacing the IDF factor with the selection functions, namely, CHI*TF, IG*TF and GAIN RATIO*TF. Other examples include TF*RF and TF*LogOR. Supervised term weighting uses class data through one of the following ways:

- Statistical confidence interval was used by Soucy and Mineau (2005).
- The weight of the terms can be calculated by using statistic feature selection measures, such as information gain, mutual information and CHI2. Based on this

Denoted by	Mathematical form	Description
Binary weight	1 Term in the document 0 Term not in the document	1 if the term appears in a document, and 0 denote the absence of the term in the document
Tf	Tf	Number of times the term occurs within the document
Log tf	$\log(1 + tf)$	Log of term frequency
TF-IDF	$w_{ij} = tf_{ij} * \log \frac{N}{df_i}$	Multiply the Tf by the inverse document frequency factor
Prob-idf	$tf * \log \left(\frac{N - n_i}{n_i} \right)$	Multiply the Tf by the probability IDF (n_i : the number of document contains t_i)

Table I.
Unsupervised term-weighting schemes

measure, the terms with a high score of this function contributes more than those with a lower value (Erenel *et al.*, 2011).

- The term weighting can be combined with the classifier. The classifier scores identify the positive documents that belong to a particular class, from negative documents that do not belong to the class. This score itself can be used as a term weight (Lan *et al.*, 2009).

Many researchers (Hemalatha *et al.*, 2013; Rosa *et al.*, 2014) used the weighting scheme for both short and long texts, without focusing on the special nature of the short text. They concentrated on the improved approach that can enhance the classification or the term selection and extraction method considering that it significantly affects the performance of the classification. Other researchers developed a scheme specialized for short texts by updating the existing scheme to meet the requirement of the short text. Quan *et al.* (2011) proposed three new supervised term-weighting schemes, namely, $qf*icf$, $qf*icf*iqf$ and vrf , for the classification of questions. The proposed methods depend on the class inverse and the term occurrences. To verify the efficiency of the three methods, an experiment was conducted on Yahoo questions as the data set using SVM and KNN. The results revealed that the $qf*icf*iqf$ method outperformed most of the term-weighting methods by achieving the best accuracy among them, and that $tf*OR$ is the best method among all the methods compared.

5. Feature selection and extraction in short texts

The main challenge to text classification is the redundancy and irrelevance of terms that form a noise to text classification. The reduction of unnecessary data can enhance the classification accuracy as it decreases the ambiguity in the whole process and reduces the complexity of the model construction by reducing the dimension of the feature space.

Most of feature selection approaches for text classification are based on term frequency (Lopez *et al.*, 2007), which refers to how many times a word occurs in a document. Other methods are based on document frequency in which the number of documents in a corpus wherein a particular word occurs are counted (Ogura *et al.*, 2009). Many researchers (Martineau *et al.*, 2008; Quan *et al.*, 2011; Ren and Sohrab, 2013) consider that the term frequency provides information about the contribution of a word in the document or in the whole corpus. However, the document frequency can only say whether a particular term is present or not in the documents.

The dimension reduction of the feature set is intended to enhance categorization performance by retaining only the important features and removing the redundant or unnecessary features. This enhancement can be achieved by using one of the two processes: feature extraction and feature selection. Feature extraction generates new features. It is the process in which the feature set transforms into a new one without removing any of the existing features. Feature selection reduces the feature space by selecting a subset of the original set without creating any new feature set. The researchers use one of these two approaches or a combination of them based on the application domain and the nature of the problem. More detailed information on approaches to feature selection are shown in Table II.

5.1 Approaches to feature selection

Three paradigms of the feature selection approaches include the filter, wrapper and the embedded approaches. The filter method uses the statistical analysis of the feature set to

choose the feature that is more discriminative. This method considers only the essential attributes of the documents, making it fast and robust against the unrelated data; it is useful when the dimension of the feature space is large. The wrapper method (Meng and Lin, 2010) is a more sophisticated feature selection method. It involves the necessary interaction with the classifier and considers performance as a criterion for feature selection. The embedded approach (Mundra and Rajapakse, 2010) is different from the wrapper method. It does not partition the data corpus into training sets and validation sets.

5.2 Filter-based approach

The filter approach applies a measure that is a static measure for assigning a score to each feature in the feature set to rank the feature and select the feature with the highly ranked score. Filter-based feature selection methods are categorized as either local or global based on whether the score assigned to the feature is a single- or multi-class score. Global feature selection attempts to choose a set of features that is discriminative among all classes. Meanwhile, local feature selection aims to find the feature that is discriminative for a certain class only. Thus, the global approach is built on the same feature set for all classes, while the local approach has a feature set for each class that contains the keywords that represent such class.

Some researchers classified the filter method as either one- or two-sided. In the one-sided filter method, the negative features are not selected because the membership feature always receives a score greater than 0, while the nonmembership feature obtains a score less than 0 (Ogura *et al.*, 2011; Martin-Bautista and Vila, 1999). If the feature is ranked in declining order, then the positive one has higher priority than the other. Relevance frequency (RF) and log-odds ratio (LOG-OR) are examples of one-sided schemes.

By contrast, the two-sided metrics include the positive and negative features for each class; both of the membership and nonmembership features obtain a score greater than or equal to 0. In two-sided schemes, the feature set infrequently contains negative features, given that the positive features obtain higher scores than negative ones (Ogura *et al.*, 2011). If the data set is balanced, then the feature set usually contains an equal mix of the positive and negative features. The symmetric correlation coefficient $|CC|$ and chi-squared (CHI) methods are examples of two-sided schemes.

5.2.1 Chi-squared (CHI2) method. This method has been extensively used in text classification; CHI2 is used to measure the dependency between a term t_k and a particular class c_j (Günel, 2012). The term t_k and class c_j are dependent when satisfying the condition in which the occurrence of term t_k the occurrence of class c_j . Consequently, the term is

References	Classifier	Method	Approach	Data set	Measure	Value
Dilrukshi and Zoysa (2014)	J48	Ratio method	Filter	Twitter corpus	Micro F1	86.0
	SVM					97.7
	Random forest					93.5
	Random tree					98.8
Wang <i>et al.</i> (2011)	simple CART	Fisher's discriminant ratio based on frequency	Filter	COAE2008 (Chinese opinion analysis evaluation)	Accuracy	82.2
	SVM					86.61
Wang <i>et al.</i> (2013)	Naïve Bayes	IDF-ICF	Filter	Web snippet	Accuracy	71.4

Table II.
Summary of some
feature selection
approaches

relevant as a feature. Thus, we can consider the term t_k an important discriminative feature to the class c_j . Moreover, it can be formulated as:

$$chi(t_j, c_i) = \frac{N[p(t_j, c_i)p(t_j^-, c_i^-) - p(t_j, c_i^-)p(t_j^-, c_i)]^2}{p(t_j)p(t_j^-)p(c_i^-)p(c_i)} \quad (1)$$

where $p(t_j)$ is the probability of the term t_j in the whole corpus; $p(t_j, c_i)$ is the probability of documents belonging to class c_i that contain the term t_j ; $p(t_j^-)$ is the probability of documents that do not contain the term t_j ; $p(t_j, c_i^-)$ is the probability of documents that do not belong to class c_i that contain the term t_j ; and $p(t_j^-, c_i)$ is the probability of documents that do not belong to class c_i that does contain the term t_j .

5.2.2 Information gain (IG) method. The IG provides a term value that represents how much the classification of the document in text mining benefits from the presence or absence of a term (Foreman, 2003). A term that is more discriminatory obtains a high score. A high score means that the document contains the specified term that belongs to a particular class. IG for term t in the class c is defined as:

$$IG(t_j) = - \sum_{i=1}^M p(c_i) \log(p(c_i)) + p(t_j) \sum_{i=1}^M p(c_i t_j) \log(p(c_i | t_j)) \\ + p(t_j^-) \sum_{i=1}^M p(c_i | t_j^-) \log(p(c_i | t_j^-)) \quad (2)$$

IG typically achieves a good performance when it used with a short text (Irani et al., 2010).

5.2.3 Correlation coefficient (CC) method. CC is considered as the asymmetric (one-sided) version of CHI2. It is based on the terms that have a high occurrence in the positive category and less occurrence in the negative category that benefits more as a discriminator to the class (Manning, 1999). As any asymmetric method, CC confers the positive terms more significance than the negative terms. It is defined as:

$$CC(t_i, c_j) = \frac{(p(t_i, c_j)p(t_i^-, c_j^-) - p(t_i, c_j^-)p(t_i^-, c_j))\sqrt{N}}{\sqrt{p(t_i)p(t_i^-)p(c_j)p(c_j^-)}} \quad (3)$$

5.2.4 Accuracy balanced (Acc2) method. This method was introduced by Galavotti et al. (2000) as a simplification method of CHI2. In Acc2, positive and negative features may be selected, as this method is classified as a two-sided approach. It takes the difference between the existence of the term in the class and all other classes to infer the importance of the term to the particular class. Acc2 is formulated as follows:

$$Acc2(t_i, c_j) = |p(t_i, c_j) - p(t_i, c_j^-)| \quad (4)$$

5.2.5 Pointwise mutual information (PMI) method. PMI measures the co-occurrence strength between two features. A high value of PMI means that the two features (words) co-occur more frequently. This metric was inferred from the information theory (Yu et al., 2013). It is used as feature selection method to determine the mutual information between the term and particular category. It is defined as the ratio between the expected co-occurrence of class C_i and term t_j . It is expressed as follows:

$$\text{PMI}(C_i, t_j) = \log \frac{F(t_j)p(C_i, t_j)}{F(t_j)p(C_i)} = \log \left(\frac{p(C_i, t_j)}{p(C_i)} \right) \quad (5)$$

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[Yu et al. \(2013\)](#) developed a new variation of the PMI by using a contextual entropy model to find a set of words from the seed to use them in classification. The expansion that uses both the co-occurrence and contextual distribution enhances the performance of the classification because of its ability to find more affected features than the basic PMI. [Ostrowski \(2014\)](#) used many feature selections on the Twitter data set. The results revealed that the PMI with the NB achieved the best result for all the matrices.

5.2.6 Odds ratio (OR) feature selection method. The difference in the term distribution in the membership and nonmembership documents measures the possibilities of the term occurrences in the positive category normalized by that of the negative category ([Foreman, 2003](#)). [Mesleh \(2011\)](#) compared the six feature selection methods based on macro-averaging F1; the experiment illustrated that the OR feature selection method outperformed all the other methods. The OR method is defined as:

$$\text{odd ratio}(t_i, c_j) = \log \frac{p(t_i, c_j)(1 - p(t_i, c_j^-))}{p(t_i, c_j^-)(1 - p(t_i, c_j))} \quad (6)$$

[Ogura et al. \(2009\)](#) used the OR, which achieved 86 per cent accuracy on short-text corpus, namely, the data set on movie reviews.

5.2.7 MOR metrics. MOR is an extended variant of the OR method. In contrast with the OR method that prefers the positive features, MOR depends on both negative and positive features. MOR was introduced for multi-class classification in which the negative term usually significantly affects categorization. MOR can be defined as:

$$\text{MOR}(t_i, c_j) = \sum_j \left| \log \frac{p(t_i, c_j)(1 - p(t_i, c_j^-))}{(p(t_i, c_j^-)1 - p(t_i, c_j))} \right| \quad (7)$$

[Chen et al. \(2009\)](#) compared MOR with other feature selection metrics, such as IG on short-text data set. The results revealed that MOR outperforms the other methods.

5.3 Wrapper and embedded approaches

The wrapper approach uses machine learning to choose the relevant subset of features depending on the prediction accuracy; thus, the classification algorithm itself is part of feature selection ([El Akadi et al., 2011](#)). The wrapper approach is generally more efficient than the filter approach, but its complexity is greater than that of the filter approach ([Unler et al., 2011](#)). The wrapper approach requires a search algorithm to find the optimal feature set. Heuristic search algorithms can be utilized as optimization methods to find the subsets of the feature that outperforms the other features. Another method using the Sequential Backward Selection begins with a set that contains all features in the corpus and gradually discards the features that are less likely to appear in the optimal set. A stop threshold point is chosen when the performance of the learning algorithms falls under a specified threshold value; the search process will then stop ([Martin-Bautista and Vila, 1999](#)). The wrapper approach usually uses the evolutionary artificial intelligence algorithm to optimize the

selection operation, such as the genetic algorithm (Tsai *et al.*, 2013), ant colony optimization (Aghdam *et al.*, 2009) and particle swarm optimization.

However, the embedded feature selection approach includes feature selection as part of the learning process. Efficiency is achieved as splitting the data to test, and validation data and training data are not needed. Each feature must also be examined, thereby requiring retraining from scratch. Consequently, the speed of finding a solution increases (Guyon and Elisseeff, 2003). As an example, Setiono and Liu (1997) used the neural network to assigning a score to the features. Multilayer networks are learned, and the features are ranked by importance. Thus, the trained network is used to compute and measure the importance and significance of the terms.

The hybrid approach combines the filter and wrapper approaches. The filter method is usually used first to obtain a set of discriminative features and eliminate noise. The dimensions of the feature space are then reduced again by applying one of the wrapper methods to obtain the best feature with less effort by reducing the computation in the wrapper method such that the method work on a small feature subset of the original space. El Akadi *et al.* (2011), Bharti and Singh (2015), Günel (2012) and Wang (2010) used the approach that combines the filter and wrapper methods in a hybrid manner to enhance the performance of categorization. Onan, A. and Korukog (2016) introduce an ensemble feature selection for text sentiment classification. The proposed approach combines the resulted subsets of several filter feature selection methods in one subset. The aggregation based on genetic algorithm, the goodness of each feature set evaluated using the Spearman foot rule distance. The aggregation strategy performs better than the individual methods. The aggregation strategy performs better than the individual methods.

5.4 Feature extraction

Feature extraction reduces the dimension of the feature space by combining or transforming the original feature set. The extraction approach aims to derive the new feature from the initial feature set with reserved the latent structure of the original data. The performance of the extraction method decreases when the feature space contains a substantial amount of irrelevant features. Partial least square (Tenenhaus *et al.*, 2005), latent semantic indexing (Deerwester *et al.*, 1990) and principal component analysis (PCA) (Kim, 2008) are examples of feature extraction.

Many researchers use PCA in text-classification applications (Zareapoor and K. R, 2015; Bharti and Singh, 2015; Veerabhadrapa and Rangarajan, 2010) because of its efficiency. As any feature extraction method, PCA transforms the original feature set to a new feature set in which its size is less than that of the initial set. In other words, PCA fabricates uncorrelated components from the original collection of correlated variables. This component represents a linear weighted integration of the original variables.

Studies on both the selection and extraction approaches in a bi-level way have been increasing; this condition benefits from both approaches, such as the hybrid approach that joins the filter and wrapper methods to maximize the accuracy of the classification. The filter method is usually used to perform the first phase of selection, followed by the application of the extraction algorithm (Bharti and Singh, 2015; Zareapoor and K. R, 2015). A two-stage method that combines both the filter feature selection and feature extraction has been proposed by Veerabhadrapa and Rangarajan (2010) to select the set of the best discriminative features. In the first stage, mutual correlation is used to reduce the dimension of the feature space, followed by the extraction of the feature using PCA or LPP. Many experiments have been conducted to assess the performance of the proposed method; the

results showed that the bi-level method outperforms the method that uses any of used approaches.

Zareapoor and K. R (2015) compared the feature extraction against feature selection used in email classification. The results revealed that feature extraction (PCA, LSA) outperforms feature selection (IG, CHI). Furthermore, LSA achieved the best accuracy. The extraction methods do not depend on the number of features chosen.

6. Short-text classification

Text classification is the process of assigning each document to a correct label of predefined classes. Many applications use text classification, such as sentiment analysis, customer review, search and many other applications, in information retrieval. Text classification is divided into two steps: training step and test step. In training, the training corpus is divided into classes, and the feature extraction will be used to eliminate all the noise and redundant terms; thus, only the relevant discriminator term remains in the data set to be used in the test step. Thus, when the test document is input, the data in the learning step will be used to assign the sufficient class label to this document.

When analyzing or classifying text, the first thing that comes to mind is the extraction of words and the computation of its frequencies to perform the classification based on the majority of the words. The other can consider the grammar and word position to perform the same operation. This condition summarizes the methods that are used in text classification, which is supervised and unsupervised. Some researchers prefer to start from the bottom by beginning in the word. The word is scored to assess its strength and the correlation between the words. Then, its subjectivity is determined, and the classification is then performed based on the subjectivity of the document. The other begins from top to bottom, starting from training a classifier using the set that is labeled documents based on their class. The familiar state of the art of both approaches with the supervised and unsupervised text classification algorithm is described briefly. As illustrated in Table III, the machine-learning algorithm becomes the focus because most of the experiments in previous years show that the machine-learning approach succeeds in short-text classification and achieves a high accuracy in many cases.

6.1 Machine learning

In machine learning, the data set needs to be divided into two sets: a training set and a test set. Training data are used to train the classifier on how to use the feature of the document

Reference	Classifier	Selection/ extraction	Data set	Measure	Value
Jiang <i>et al.</i> (2011)	SVM	PMI	Self-collection using Twitter API	Accuracy	68.3
Tsuchida <i>et al.</i> (2012)	NN training	POS	Self-collection using Twitter API	Accuracy	68
Weissbock <i>et al.</i> (2013)	SVM, Naïve Bayes, DT	Use the NTLK	Self-collection using Twitter API	F measure	88
Wang <i>et al.</i> (2013)	Naïve Bayes	IDF-ICF	Web snippet	Accuracy	71.4
Rosa <i>et al.</i> (2014)	Fuzzy Fp KNN SVM	TF_IDF	Self-collection using Twitter API	F1 measure	84 27 74

Table III.
Summary of some machine-learning algorithm articles

within these data sets to distinguish the documents from each other, whereas a test set is used to validate and evaluate the performance of the classifier. Figure 2 illustrates the different machine-learning approaches. Two types of machine-learning algorithms are unsupervised and supervised learning. The latter requires a large amount of labeled documents from the training corpus to achieve learning, whereas the former categorizes the example without the need for further knowledge. Sometimes, some problems with labeled data are not available. Instead, similar documents are grouped in clusters. Clustering can be done using two approaches: hierarchical clustering (such as the agglomerative hierarchical clustering) and partitional clustering (such as the K-means clustering). However, sometimes collected labeled data require plenty of efforts due the data sources or the annotation of these data may be inaccurate. In such cases, researchers use semi-supervised learning approaches where there are a little number of labeled data, and the rest are not labeled (Yin *et al.*, 2015). Here, the focus is on the supervised approach because of the prevalence and high efficiency on text classification.

6.1.1 Example-based classifier. Instead of construct class description from the training document, this method is based on a measure of the similarity of the testing document and every document in the training corpus. It simply keeps the related training document implemented in a specific representation, like TF under the same class (Feldman and Sanger, 2007). However, this classification decision-making process takes a long time. An example of this method is the KNN classifier, which is a straightforward classifier. It aims to find the K documents that obtain the maximum similarity with the new document. It measures the similarity between the documents by using the cosine measure, the Euclidean distance or another similarity formula. The K documents that have the greatest similarity with the presented document are then selected according to the highest similarity class assigned to the appropriate label with the tested document. However, for each new test document, the similarity with all the documents in the data set needs to be found.

6.1.2 Probabilistic classifier. This classifier is regarded as a generative classifier. Probabilistic classifiers assume that each category is a component of the combination. The most appropriate class that the document should belong to not only needs to be found, but it

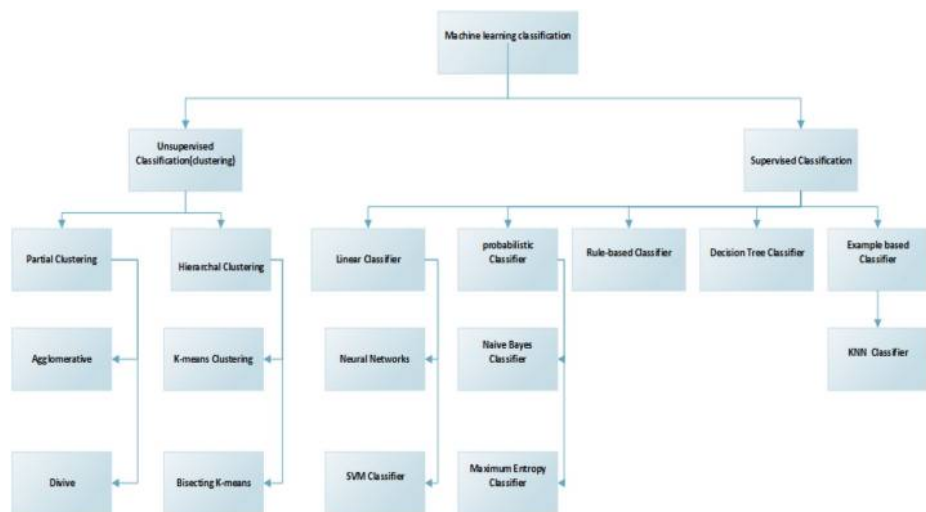


Figure 2.
The machine-learning classifiers

is also able to predict the probability distribution of a given document among a set of classes. The NB and Max entropy are examples of probabilistic classifiers.

One of the simplest and oldest classifiers is the NB. It finds the probability of the document that belongs to a particular class by combining the probability of all features in the class. The classifier estimates the posterior probability of a category based on the word occurrence. It works properly with the BOW's model, which assumes that all the words are independent and does not regard the position of this word in its document. Through the use of Bayes theory, the probability of a given the word w_i to belong to a particular class c_k can be formulated as follows:

$$p(c_k, w_i) = \frac{p(w_i|c_k)p(c_k)}{p(w_i)} \quad (8)$$

By calculating the probability of all terms for every class, the vector probabilities ($p(c_k|t_1)$, $p(c_k|t_2)$, ..., $p(c_k|w_i)$) are built where $p(w_i|c_k)$ is the probability of the term t_i in c_k . The probabilities of the documents in the class are then computed. For the document to be classified, its probabilities need to be computed and compared with the probability of the documents in every class, and then classified with the document to the class with the most similarity. The probability of an input document to be calculated is as follows:

$$p(c_k, w_i) = \frac{p(c_k|w_1, w_2, \dots, w_n)}{\text{the total number of a word extracted from the document } d} \quad (9)$$

A reasonable performance can be achieved with a short text (Kang *et al.*, 2012; Lane *et al.*, 2012). In the experiment of Irani *et al.* (2010), to classify tweets to its trends topic, they observed that utilizing NB with 100 features would be ideal for a hybrid data set combined with the tweets and their associated web pages.

Maximum entropy (ME) classifier is a probabilistic exponential classifier that assumes that the features are dependent. ME is a generalization of the paradigm utilized by NB. ME is commonly used when the prior distributions of the data is unknown. This classifier calculates the weight for each feature that maximizes the probability of the training data. Thus, for each feature, it finds its joint feature. The weight of each joint feature is then computed by adopting iterative optimization, such as quasi-Newton optimization and iterative scaling to maximize the log likelihood of the training data (Zhu *et al.*, 2005).

6.1.3 Decision-tree classifier (DT). A DT classifier is extensively used in data mining, as it is easy to understand. A tree that represents the presented model needs to be built; the training data space is breakdown in a hierarchical representation for the restriction on the node value to divide the data. The leaves node indicated the variable target value labeled by the categories label, and the inner node and root node that represent the input values, which are labeled by the features, exist. A DT classifies the documents by moving down through the branch whose restrictions are satisfied by the document from the starting point to the root node toward the edge node that represents the category until reaching the leaf whose label is suitable to be assigned to the desired document. Training is done by partitioning the feature space into subsets by its predefined feature value in text classification. Splitting depends on the presence or absence of words or phrases.

The most used DT algorithm for text classification is C4.5 and its successor variants C5. This algorithm achieved a reasonable performance (Gunal, 2012; Uguz, 2011). In study of Irani *et al.* (2010), the C4.5 classifier obtains the highest F1 value when compared with NB

and stump classifiers by applying them on the enrichment of the data set that consists of tweets and their associated web pages to classify the tweets to their trending topics.

6.1.4 Linear classifier. In the linear classifier, decision-making is based on the linear combination value of the features. These types of classifiers usually achieve a good result, especially in text categorization. Two of the most popular linear classifiers are presented in the next subsections.

SVM is a statistical classifier proposed by Vapnik ([Hsu and Lin, 2002](#)), which depends on the structural risk minimization principle. It separates the learning data with maximum margins. The hyperplane needs to be found in the search space that separates the different classes in the training data. It achieves the separation that represents the maximum margin of the data points. It finds the decision surface that separates the member and nonmember of the training data of a particular class with the highest margin. The best surface takes the effective components in the training data and removes all the other data. SVM mainly deals with a binary class problem. However, it can be used in a multi-class problem by dividing the problem into a subproblem ([Isa et al., 2008](#)).

As it achieves high performance, it is considered one of the best classifiers, especially in text classification ([Soucy and Mineau, 2003](#); [Taşcı and Güngör, 2013](#)). SVM is also considered one of the ideal solutions in text classification because of the text data described as sparse data in which the text tends to be correlated and scattered into linearly separated classes ([Hsu and Lin, 2002](#)). [Hsu and Lin \(2002\)](#) suggested two approaches to using SVM. In the multi-class classification, namely, one-against-all and one-against-one ([Günel, 2012](#)), [Ren and Sohrab \(2013\)](#) used SVM with other classifiers. SVM generally outperforms the other classifiers in most cases.

Neural network (NNs) are another linear classifier example. The neurone is the essential component of the neural network, which consists of many neurones joined by weighted links. The feature value is loaded to the input node. The links weight indicate the relation and dependencies to the output node, which represent the classification value.

The multi-layer NN is used in both linear and non-linear classification problems. Moreover, it consists of three parts: input, output and hidden layers. Using one hidden layer is sufficient for many situations, whereas in other cases, using two hidden layers may enhance the performance ([Fausett Laurene, 1994](#)). The input nodes receive the vector X , which is the feature weights of the documents in the training data propagated through the network of neurones; the class label is determined by the output neurones. The basic type of NN is the linear classifier, which consists of only two layers: the input and the output; this type is known as a perceptron. However, in multi-layer NNs, the error needs to be back-propagated through various layers, and this requirement further complicates the training process. Thus, the back-propagation approach is one of the most efficient learning methods. The input of each layer is the output of the previous layer. The learning process starts by loading the document in the training set to the entry node to compute the net inputs, which will be used by the transfer function (e.g. logarithmic sigmoid function) to find the output. The second step is that the error is back-propagated through the network, and the weights are adjusted to reduce the error ([Feldman and Sanger, 2007](#)). This process is repeated until a stopping criterion is reached, which may be a predefined repetition count or the lowest error goal.

[Moraes et al. \(2013\)](#) conducted an empirical experiment to compare SVM and the NNs. Although SVM is used more often than the NN in text classification, the NN achieves better accuracy than the SVM. The experimental results revealed that the computational cost decreases when IG is adopted for both methods.

7. Ensemble classification

Each classifier generally has its strengths and weaknesses that affect its performance. An ensemble will combine the strengths of multiple classifiers to maximize predictive performance. It can be a way to avoid the drawback of some methods or to benefit from a set of classifiers so that the risk of making a wrong decision is avoided. Moreover, bias classification can be avoided (Wang *et al.*, 2014). The ensemble classification has a good performance and accuracy (da Silva *et al.*, 2014). The performance of the ensemble approach depends on the combination of the base-learning algorithms, as well the selection of the appropriate combination method for base-learning algorithms.

Two ensemble approaches are typically used in classification: dependent and independent approaches (Rokach, 2010). In the independent method, the individual classifiers are learned using the same training corpus at the same time; the results are then combined, and the best result is selected among all the classifiers (Feldman and Sanger, 2007). Aggregation of the results of the different classifiers can be done in different ways. The most popular one adopts the majority rule in the result of the ensemble classifier and selects the best viewing result. Bagging and dagging are examples of independent methods, whereas the dependent methods improve the categorization by training many classifiers sequentially. Before learning the next classifier, the training corpus is reweighed with a maximum weight assigned to the documents that were misclassified by the preceding classifiers. One of the examples of a dependent approach is the boosting algorithm (Hassan *et al.*, 2013).

Boosting algorithms function by boosting iteratively the updating of the weight of instances in the training set so that the weight of the misclassified instances is increased, whereas the weight of the classified instances are reduced. AdaBoost is an example of this technique (Hassan *et al.*, 2013), Ammar Hassan proposed the bootstrap ensemble method for sentiment analysis in Twitter (BPEF). Seven different classifiers were used, namely, SVM, NB, bays net, random tree, NN, logistic regression and REP tree. The experimental results revealed that the BPEF obtained the highest score for recall compared with other algorithms and techniques tested in the experiment.

Bagging (bootstrap) the result obtained by different base-learning algorithms that run on different training data subsets that are aggregated using majority voting. Given that these subsets were built randomly from the main training data set, each of them contains a relatively sufficient amount that considers the overlap with other subsets. A tweets sentiment approach that is based on lexicon-based classification was proposed by da Silva *et al.* (2014). Four classifiers, namely, SVM, logistic regression, NB and random forest, were combined using the bagging approach. The sentiment result was obtained from the four classifiers by combining the results of different classifiers using the majority voting. The results showed that the ensemble method enhances the classification accuracy. When using the BOW and lexical opinion features together, an accuracy of about 80 per cent is achieved. More details on ensemble classification models in the short text are summarized in Table IV.

Another independent method is the dagging algorithm. The bagging method is for the bootstrapping of training data, whereas the dagging method creates a disjointed and stratified chunk of data and supplies it to the base learner. The results of different learners are combined using the majority vote (Kotsianti and Kanellopoulos, 2007). Other methods for selecting the training samples are stack and random space algorithms (Onan *et al.*, 2016).

Reference	Classifier	Selection/ extraction	Data set	Ensemble method	Measure	Value
Hassan <i>et al.</i> (2013)	NN	Hybrid	Telco tweets Tech tweets Pharma tweets	Bootstrap	Accuracy	71.7
	Random Tree					76.3
	REP tree					67.8
	Naïve Bayes					
	Bayes net					
Da Silva <i>et al.</i> (2014)	Logistic regression	Bag-of-words and lexicon	Sanders Twitter Stanford Twitter Obama-McCain Debate Health care reform	Majority vote	Accuracy	82.7
	SVM					85
	Multinomial					76.8
	Naïve Bayes					75.1
	SVM					
	Random forest					
	Logistic regression					
Lin and Kolcz (2012)	Logistic regression	Feature hashing	Self-collection Using Twitter API	Majority vote	Accuracy	82
	Logistic regression					
Tuarob <i>et al.</i> (2014)	MNB	N-gram LDA Dictionary based	Twitter B Facebook data set	Multi-Staging reverse multi- staging Majority vote weight probability averaging		46.20
	multinomial					47.07
	Naïve Bayes					
	Naïve Bayes					
	Random forest					
Example ensemble classification models	Ripper					
	SVM					

Table IV.
Example ensemble
classification models

8. Classification evaluation measures

As a rule, the effectiveness of the evaluation of short-text classification is tested experimentally and assessed using standard measures, including recall, precision F_1 , accuracy and error rate. Suppose that we have a class c_i , is one of the class in C such that $C = \{c_1, c_2, \dots, c_k\}$ is the set of classes. At the beginning, the following values for each class c_i that represent the primitive factor are computed to calculate all other different measures:

- True positives (TP_i): The number of documents in the test corpus that belong to class c_i and are classified to be belonging to class c_i .
- True negatives (TN_i): The number of documents in the test corpus that belong to class c_i and are classified not to belong to class c_i .
- False positives (FP_i): The number of documents in the test corpus that do not belong to class c_i and are classified to be belonging to class c_i .
- False negatives (FN_i): The number of documents in the test corpus that do not belong to class c_i and are classified not to belong to class c_i .

The recall $R(c_i)$ for a given class c_i is defined as the portion of the tested document that are classified to class c_i to the test documents of class c_i :

$$R(c_i) = \frac{TP_i}{TP_i + FN_i} \quad (10)$$

The precision $p(c_i)$ for a given class c_i is defined as the portion of the tested documents that are classified to class c_i from all the documents that belong to class c_i :

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$$P(c_i) = \frac{Tp_i}{Tp_i + Fp_i} \quad (11)$$

The global recall $RG(c_i)$ is defined as:

$$RG(c_i) = \frac{\sum_{i=1}^k Tp_i}{\sum_{i=1}^k (Tp_i + Tn_i)} \quad (12)$$

The global precision $PG(c_i)$ is defined as:

$$PG(c_i) = \frac{\sum_{i=1}^k Tp_i}{\sum_{i=1}^k (Tp_i + Fp_i)} \quad (13)$$

A usual approach to evaluating the classification effectiveness is F_1 , which is a combination of precision and recall given by their harmonic mean, is defined mathematically as:

$$F1 = \frac{2P(c_i)R(c_i)}{P(c_i) + R(c_i)} \quad (14)$$

where R and P are the recall and precision values, respectively. Two standard $F1$ scores are used to evaluate the results of categorization. The first one is micro- $F1$, which is the $F1$ measure that is computed over global recall and global precision:

$$\text{micro} - F1 = \frac{2PG(c_i)RG(c_i)}{PG(c_i) + RG(c_i)} \quad (15)$$

The second score is the macro-average $F1$, which is the average of the $F1$ values for each class:

$$\text{macro} - F1 = \frac{\sum_{i=1}^k \left(\frac{2PG(c_i)RG(c_i)}{PG(c_i) + RG(c_i)} c_i \right)}{K} \quad (16)$$

Micro- $F1$ tends to be dominated by the classifier's performance on common categories, and the macro- $F1$ is more influenced by the performance of rare categories (Fiaidhi *et al.*, 2013). If the classifier predicted none of the documents to a given class, its $F1$ is defined equal to 0.

The accuracy is defined as the overall accuracy of the result of the text classification and is considered as the addition of the classification parameters (Tp , Tn) divided by the entire number of text with classification parameters ($Tp + Tn + Fp + Fn$).

$$\text{accuracy} = \frac{Tp + Tn}{Tp + Tn + Fp + Fn} \quad (17)$$

The error rate is the percentage of documents that are assigned to the wrong class:

$$\text{Err} - \text{rate} = 100\% - \text{accuracy} \quad (18)$$

9. Evaluation and discussion

When developing a framework for short-text classification, knowing the previously used strategies and methods is necessary. As such, we presented an overview of the machine-learning algorithms that have been used and the results of some studies in this domain. In many cases, the performance of various approaches was evaluated by calculating accuracy or F-measure metric. Other metrics, such as precision, recall and error rates, provide significance data, but are rarely used. Precision is the portion of tested instances that are classified properly, whereas recall is the portion of tested document belonging to a specific class. The error rate denotes the rate of misclassified instances. Figure 3 shows the comparison between a sample of published works on short-text classification in terms of achieved accuracy, where the vertical axes represent the average achieved accuracy. For more details, Table V summarizes the references used in this study. High accuracy was achieved by most publications as a result of using a self-building data set, most of which were collected by Twitter API. High accuracy may be attributed to the use of unbiased data because those publications that used to benchmark data set obtained low accuracy.

The sparsity of feature space is a challenge in short-text classification. The vast number of features disperses the classifier and leads to a poor result. Using feature selection leads to improved performance, especially for the short-text domain where data have high sparsity and great amount of noise. Figure 4 shows that most of the publications use feature selection methods to clean the feature space. Moreover, the filter methods are considered as the most common for their simplicity. Most of the studies were attracted toward the supervised approach, while a few explored the unsupervised one. This aspect is due to the availability of labeled data if microblogs and reviewers are considered as the main sources of short text

Figure 5 shows the overall distribution of the classification approaches used in the studies. Clearly, Most of the research has used the linear base classifier, while not much researchers used the example-based approach. The probabilistic base came second, where 21 per cent of the researchers prefer. We can see that the ensemble and DT are relatively close to each other.

We cannot include all studies in one collection, because it is difficult to find enough number of research used the same algorithms, data and apply the same experimental conditions. Tables II-V illustrate the results of all studies used in this article, the experiments performed on different data sets, most of the studies have focused on Twitter, while other

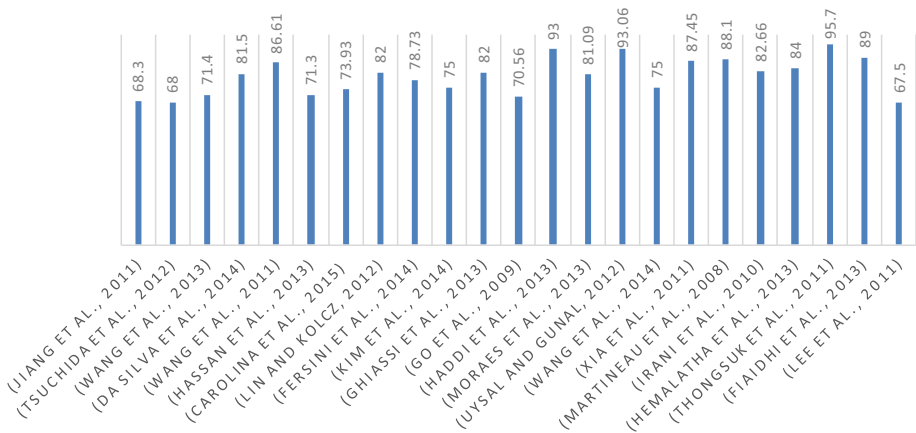


Figure 3.
Comparative results
among different
literature in terms of
achieved accuracy

Reference	Classifier	Selection	Weights	Data set	Area	Measure	Value
Jiang <i>et al.</i> (2011)	SVM	PMI	Binary 0,1	Self-collection using Twitter API	Sent	Accuracy	68.3
Tsuchida <i>et al.</i> (2012)	NN training	POS	Binary 0,1	Self-collection using Twitter API	Class	Accuracy	68
Weissbock <i>et al.</i> (2013)	SVM, Naive Bayes, DT	NLTK		Self-collection using Twitter API	Class	F measure	88
Wang <i>et al.</i> (2013)	MAXENT	TFIDF	DFICF	Web snippet	Class	Accuracy	71.4
Dilrukshi and Zoyza (2014)	J48 SVM Random forest Random tree Simple CART SVM	Ratio method		Twitter corpus		Micro F1	94
Wang <i>et al.</i> (2011)		Fisher's discriminant ratio based on frequency	TFIDF	COAE2008s	Sent	Accuracy	86.61
Da Silva <i>et al.</i> (2014)	ENS(LR + RF + MNB)	TF	TF	Sanders Twitter Stanford Twitter Obama-McCain Debate Health care reform Telco tweets Tech tweets Pharma tweets	Sent	Accuracy	82.7 85 76.8 75.1
Hassan <i>et al.</i> (2013)	ENS (RBF NN, random tree, REP tree, Naive Bayes, Bayes net, logistic regression and SVM)	TF	TF		Sent	Accuracy	71.7 76.3 67.8
Lin and Kolcz (2012)	ENS (logistic regression)	Feature hashing	Binary	Private data set	Sent	Accuracy	82
Tuorob <i>et al.</i> (2014)	ENS (MNB multinomial Naive Bayes Random forest Ripper SVM)	Dictionary-based compound features (DC)	Binary, frequency and TF-IDF	Self-collection using Twitter API Facebook data set	Sent	F measure	46.20 47.07

(continued)

Table V.
Summary of articles
used in this study

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Table V.

Reference	Classifier	Selection	Weights	Data set	Area	Measure	Value
Carolina <i>et al.</i> (2015)	Naive Bayes SVM J48 KNN		TFIDF	OMD, SANDERS, SS-Twitter, SemEval	Sent	Accuracy	77.01 64.25 76.18 78.3
Vidya <i>et al.</i> (2015)	SVM Naive Bayes DT	POS	–	Self-collection using Twitter API	Sent	F measure	89.03 89.8 84.43
Ghiassi <i>et al.</i> (2013)	DAN SVM		TFIDF	Self-collection using Twitter API	Sent	Accuracy	80.35 78.1
Go <i>et al.</i> (2009)	SVM NN	PCA	TF	Sanders	Sent	Accuracy	75.4 67.4 68.9
Montejo-Ráez <i>et al.</i> (2014)	Naive Bayes SVM		TFIDF	Self-collection using Twitter API	Sent	F measure	62.85
Balahur and Perea-ortega (2015)	SVM			SemEval 2013 TASS 2013	Sent	F measure	53
Fersini <i>et al.</i> (2014)	ENS (SVM, DIC, Naive Bayes CRF)		Binary, TF, TF-IDF	MovieData ProductData ProductDataMD	Sent	Accuracy	79.2 75.5 81.5
Haddi <i>et al.</i> (2013)	SVM	Chi2	TFIDF, FF, FP	Movie review	Class	Accuracy	93
Kim <i>et al.</i> (2014)	SVM		TFIDF TF	Open directory project (ODP), Daum directory	Class	Accuracy	79 75
Goyal (2015)	KNN DT ELM		TF	Self-collection using Twitter API		F measure	98 100 93.89
Tripathy <i>et al.</i> (2014)	Naive Bayes SVM		TFIDF	Movie review	Sent	Accuracy	89.5 94
Moraes <i>et al.</i> (2013)	Naive Bayes SVM NN	IG CHI2	Binary, TF, TF-IDf	Movie review GPS, Books and Cameras		Accuracy	73.75 84.77 84.75

(continued)

Reference	Classifier	Selection	Weights	Data set	Area	Measure	Value
Wang <i>et al.</i> (2014)	Ensemble		TFIDF	Review data set	Sent	Accuracy	75
Quan <i>et al.</i> (2011)	SVM		Q ⁺ -icf	Yahoo question	Class	F measure	37.10
Uysal and Gunal (2012)	KNN		Q ⁺ -icf* ⁺ lqfvr ⁺ f				47.37
	SVM	IG, GI, CHI2, DP, DFS		SMS	Class	Accuracy	91.48
	NN						94.3
	DT						93.4
Xia <i>et al.</i> (2011)	ENSEMBLE		TFIDF	Cornell movie-review	Class	Accuracy	87.45
Martineau <i>et al.</i> (2008)	SVM		Delta-TFIDF	Movie review	Sent	Accuracy	88.1
Lee <i>et al.</i> (2011)	Naïve Bayes		TF_IDF	Self-collection using Twitter API	Class	Accuracy	65
	C5.0						70
Rosa <i>et al.</i> (2014)	Fuzzy FP		TF_IDF	Self-collection using Twitter API	Sent	F measure	84
	KNN						27
	SVM						74
Hemalatha <i>et al.</i> (2013)	Naïve Bayes		TF_IDF	Self-collection using Twitter API	Sent	Accuracy	84
Thongsuk <i>et al.</i> (2011)	SVM	LDA		Self-collection using Twitter API	Class	Accuracy	95.7
Irani <i>et al.</i> (2010)	Naïve Bayes, C4.5, decision stumps	Information gain		Self-collection using Twitter API	Class	Accuracy	77
							90
							81
Fiadhi <i>et al.</i> (2013)	KNN, Naïve Bayes, LM		TFIDF	Self-collection using Twitter API	Class	Accuracy	91
							88
							88

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Table V.

Figure 4.
Distribution of
feature selection
approaches

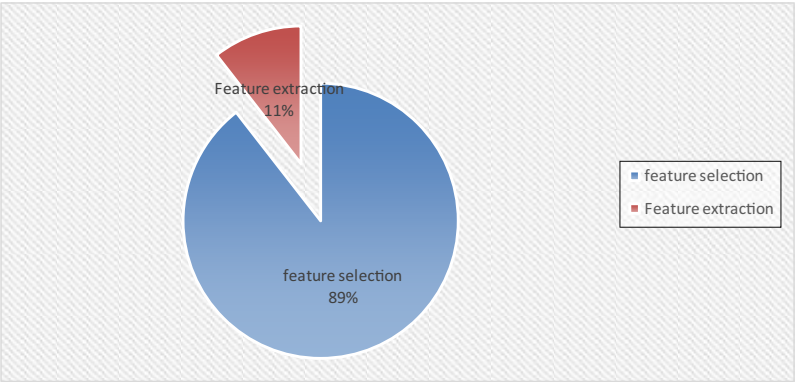
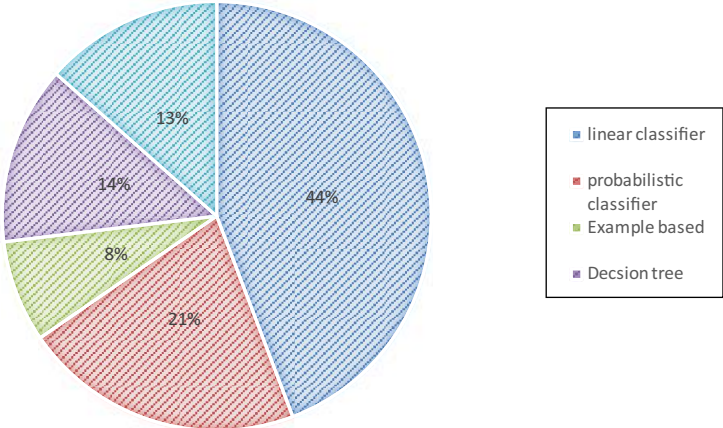


Figure 5.
Distribution of
classification
algorithms



data sets are used for sentiment related to a specific product. There are a large number of machine-learning approaches available; current research cover the most common supervised approaches which are example base, DT, probabilistic, linear and ensemble approach. While linear algorithms generally offered the best performance, the comparison between classification algorithms can only be done using the same data set because each algorithm has different behavior for different data. The best classifier used by each study is shown in [Table V](#).

10. Conclusion

Short-text classification is considered as a complex problem, as millions of daily posts (short text) in various topics are produced. The redundancy and noise in data are a common problem in social media and in different applications that use short text. This paper presented an overview of the latest techniques in short-text classification and its applications. The existing solutions, as well as their pros and cons, were summarized. The summary starts with the identification of short text and the challenge it poses to the text classification domain, followed by the depiction of the limitations of preprocessing in

cleaning this type of data, transforming it as an input for different machine-learning classifications. We emphasized on the state-of-the-art weighting schemes and feature selection and presented the approaches that can be used in short and long texts. The best evaluation measure that can be used to assess each approach is presented in the last section.

The comparative analysis confirmed the importance of various approaches and impact of using different methods on different data sets. The analysis and comparison focus on the supervised approach because it is the most common. Therefore, there is a need to explore the semi-supervised and unsupervised approaches to solve the problem of short-text classification. It was also noted, some methods produced a high performance in some studies but low performance with others. From the above analysis, it is evident that neither methods consistently beats the other. It was also observed that the preprocessing of the data combined with feature selection can yield better results.

Using a powerful short-text classification method significantly affects many applications in terms of efficiency enhancement. Current solutions still have low performance, implying the need for improvement. This review discusses related issues and approaches to these problems. Further research could be conducted to address the challenges in short texts and avoid poor accuracy in classification. Problems in low performance can be solved by using optimized solutions, such as genetic algorithms that are powerful in enhancing the quality of selected features. Soft computing solution has a fuzzy logic that makes short-text problems a promising area of research.

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Further reading

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